
Question #: 1

A 59-year-old man comes to the emergency department because of a 2-hour history of a fast, irregular heartbeat. He is diaphoretic and in respiratory distress. Physical examination shows tachycardia (increased pulse rate). In which of the following structures are the postganglionic cell bodies of the nerves fibers responsible for his sweating, most likely located?

- A. Lateral horn S2-S4
- B. Lateral horn T1-L2
- ✓C. Paravertebral ganglion
- D. Preaortic ganglion
- E. Ventral horn

Rationale: Sweating is a sympathetic response. Postganglionic sympathetic cell bodies going to the sweat glands in the body wall are found in the sympathetic chain, which is also called the paravertebral ganglion.

Question #: 2

A 66-year-old man comes to the physician because of the sudden onset of severe lower back pain. The pain started while he was bending to pick up a bag of soil. Physical examination shows loss of sensation and muscle weakness in his left lower limb. MRI shows a rupture of the intervertebral disc between L5 and S1 with herniation of the nucleus pulposus. Which of the following nerves is most likely impinged?

- A. L2
- B. L3
- C. L4
- D. L5
- ✓E. S1

Rationale: The nerve root that exits between the L5 and S1 vertebrae is L5, however the nerve root that will be compressed is S1. This is because the L5 nerve root exits above the intervertebral disc and so is not impinged.

Question #: 3

A 21-year-old man comes to the emergency department because of a 4-hour history of severe headache and neck stiffness. Meningitis is suspected and a lumbar puncture is performed to extract cerebrospinal fluid for testing. Between which of the following two vertebrae should the

needle most likely be inserted when performing this procedure?

- A. T11-T12
- B. T12-L1
- C. L1-L2
- D. L2-L3
- ✓E. L4-L5

Rationale: In an adult, the spinal cord ends at the L1/L2 level. Therefore, in order to avoid damaging the spinal cord when performing a lumbar puncture, your needle needs to be inserted below the L2 level.

Question #: 4

A 22-year-old woman comes to the physician because of a 1-hour history of a deep cut over the back of her shoulder. Physical examination shows an infected laceration over the spine of her scapula. Into which of the following lymph nodes will the infection most likely initially drain?

- A. Pectoral/anterior
- ✓B. Subscapular/posterior
- C. Central
- D. Humeral/lateral
- E. Apical

Rationale: The scapular(posterior) group of the axillary lymph nodes drains the posterior shoulder and back. The pectoral nodes drains the anterior chest wall and breast, while the humeral nodes drains the medial aspect of the upper limb. Lymphatic fluid then drains to the central and then the apical lymph nodes.

Question #: 5

A 65-year-old man comes to the physician because of a 1-month history of weakness in his back when trying to lift things from the ground. He also has difficulty keeping an upright posture while standing. Electromyography shows that the middle muscle of the erector spinae group is weakened. Which of the following is most likely the weakened muscle?

- A. Iliocostalis
- ✓B. Longissimus
- C. Multifidus
- D. Spinalis

E. Semispinalis

Rationale: The erector spinae group of muscles form the intermediate group of the deep back muscles. They are responsible for maintaining posture. The iliocostalis is the most lateral, the longissimus lies in the middle and the spinalis is the most medial.

Question #: 6

A 15-year-old girl is brought to the physician by her mother for a routine examination. Physical examination shows she is unable to stand with her back straight against the wall while her height is being measured. An x-ray of her spine shows a significant increase in the thoracic curvature. Which of the following terms best describes this deformity?

- A. Lordosis
- B. Scoliosis
- C. Spondylolysis
- ✓D. Kyphosis
- E. Spondylolisthesis

Rationale: Kyphosis describes an increase of the primary curvature of the thoracic region of the back.

Question #: 7

A 25-year-old woman is admitted to the hospital being involved in a motor vehicle collision that caused violent hyperextension of her neck and rupture of the C5/6 intervertebral disc. The nucleus pulposus of the disc has herniated through the annulus posterolaterally and is impinging upon the C6 nerve root. This is causing pain in her upper extremity. Which of the following ligaments most likely prevents the herniation from impinging upon the spinal cord?

- A. Supraspinous
- ✓B. Posterior longitudinal
- C. Anterior longitudinal
- D. Ligamentum flavum
- E. Ligamentum nuchae

Rationale: The posterior longitudinal ligament lies on the posterior aspect of the body of the vertebrae and the IV disc. It will prevent the herniation from impinging on the spinal cord.

Question #: 8

A 35-year-old man comes to the emergency department with a 2-day history of a swollen and erythematous right thumb following a deep puncture wound he sustained while working with untreated wood. His temperature is 38.3°C (100.9°F). Physical examination shows redness, warmth, and pain in the affected area. The initial management includes wound care and antibiotic therapy. Into which lymph nodes is the infection most likely to initially drain?

- A. Pectoral/anterior
- B. Subscapular/posterior
- C. Central
- ✓D. Humeral/lateral
- E. Apical

Rationale: Initial lymphatics from a deep wound would go to the lateral group, because all deep lymphatics from the upper limbs drain into the lateral group.

Superficial lymphatics also drain to the lateral group except for the lateral hand, arm and forearm, which drain into the apical group.

Question #: 9

A 1-year-old girl is brought to the physician **by her parents** for a follow-up examination . She has a fluid-filled swelling over the lower lumbar vertebrae. An x-ray of the spine shows non-union of the vertebral arches of the L4 and L5 vertebrae. MRI shows the spinal cord in the normal position but the meninges are protruding out into the swelling. Which of the following is the most likely diagnosis?

- A. Spina bifida cystica with meningocele
- B. Spina bifida with myeloschisis
- ✓C. Spina bifida cystica with meningocele
- D. Spina bifida occulta
- E. Sacrococcygeal teratoma

Rationale: The patient has spina bifida cystica, as seen by the cystic swelling in the lumbar region. Because the swelling contains meninges, but does not contain the spinal cord, the condition is spina bifida cystica with meningocele.

Question #: 10

A 55-year-old woman comes to the physician for a follow-up examination after suffering a deep laceration to the right side of her neck. Physical examination of the right shoulder shows drooping and weakness in shrugging. Which of the following nerves is most likely damaged?

- A. Thoracodorsal
- ✓B. Spinal accessory
- C. Dorsal rami
- D. Intercostal nerves
- E. Axillary

Rationale: The trapezius muscle elevates the scapula and therefore the shoulder. It is innervated by the spinal accessory nerve. Damage to this nerve will result in difficulty in shrugging the shoulder.

Question #: 11

A 25-year-old man comes to the emergency department because of a 12-hour history of severe vomiting and diarrhea. He is diagnosed with gastroenteritis which includes inflammation of the epithelial lining of the gut. From which of the following germ layers is this lining most likely derived?

- ✓A. Endoderm
- B. Paraxial mesoderm
- C. Intermediate mesoderm
- D. Lateral mesoderm
- E. Ectoderm

Rationale: The epithelium lining of the gut is derived from endoderm.

Question #: 12

A 59-year-old man comes to the physician because of a 1-week history of intermittent abdominal pain. Physical examination shows tenderness on palpation over the right costal margin. A diagnosis of inflammation of the gallbladder (root –cholecyst -) is made. Which of the following suffixes when added to the root word best describes his condition?

- A. Cyte
- B. Lith
- C. Logy

D. Crine

✓E. Itis

Rationale: Suffixes are added to the END of words root or combining forms to modify its meaning. The suffix itis, when added to a root word indicates inflammation.

Question #: 13

A 12-year-old boy is brought to the physician by his mother because of a 4-hour history of swelling on his arm after he was bitten by a mosquito. Physical examination shows a raised and reddened papule on his left arm. Which of the following root words best describes the color of the swelling?

A. Leuk/o

B. Cyan/o

C. Poli/o

✓D. Erythr/o

E. Melan/o

Rationale: Leuk/o: "white"

Cyan/o: "blue"

Poli/o: "gray,"

Erythr/o: Correct. "red"

Melan/o: "black"

Question #: 14

A radiologist would like to take an image of the body in a plane parallel to both scapulae. Which of the following best describes the desired sectioning?

A. Transverse section

✓B. Coronal section

C. Sagittal section

D. Horizontal section

E. Cross section

Rationale: The coronal plane divides the body into anterior and posterior halves.

Question #: 15

A 56-year-old man with type 2 diabetes mellitus comes to the physician because of chronic ulceration of the skin immediately superior to his left ankle. The ulcer was first treated 6 months ago and has become gangrenous. Amputation of the left leg with knee sparing is scheduled. The most likely area for the amputation cut is demonstrated by which of the following areas on the attached image?

- A. A
- B. B
- C. C
- ✓D. D
- E. E

Rationale: A and C = Left side (excluded).

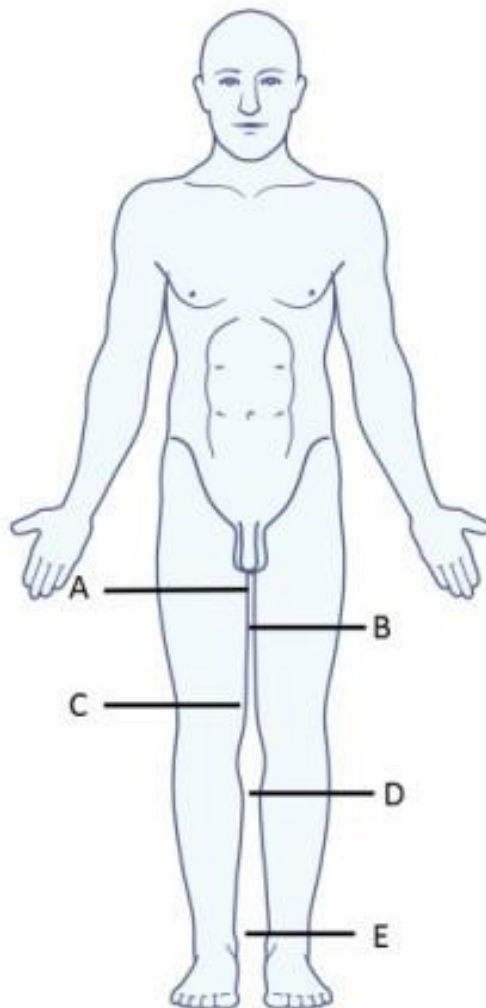
B, D, and E = Right side (possible).

B Above-knee Amputation = (doesn't spare the knee)

D Below-knee Amputation that spares the knee and is high enough to clear the gangrene. (correct)

E Foot-Amputation: Spares the knee but doesn't clear the gangrene

Attachment:



Question #: 16

A 70-year-old man comes to the physician because of a 6-month history of a lump in his abdomen which has gradually increased in size. Physical examination shows a large pulsatile mass palpated on the anterior abdomen. A CT scan of the abdomen shows an abdominal aortic aneurysm as seen in the attached image. Which of the following best describes the plane of section used in this scan?

- A. Coronal section, dividing the body into superior and inferior parts
- B. Sagittal section, dividing the body into anterior and posterior parts
- C. Axial section, dividing the body into left and right parts
- D. Axial section, dividing the body into anterior and posterior parts
- ✓E. Sagittal section, dividing the body into left and right parts

Rationale: The image is of a sagittal CT scan. This divides the body into left and right halves.

Attachment:



AAA completion.jpg

Question #: 17

A 16-year-old boy is brought to the emergency department by his mother because of a 2-hour history of severe pain in his upper limb after falling. Physical examination shows a swollen, elbow joint with limited range of motion. He refuses to let the elbow be touched due to severe pain. Which of the following actions will most likely be affected?

- ✓A. Extension
- B. Abduction
- C. Adduction
- D. Medial rotation
- E. Lateral rotation

Rationale: The elbow joint has two types of joint combined:

1. Hinge joint that permits flexion and extension, and 2. pivot joint that permits pronation and supination.

Question #: 18

A 66-year-old man comes to the physician because of the sudden onset of severe lower back pain. The pain started while he was bending to pick up a bag of soil. Physical examination shows loss of sensation and muscle weakness in his left lower limb. MRI shows rupture of the intervertebral disc between L5 and S1 with herniation of the nucleus pulposus. Which of the following embryological structures is most likely responsible for the formation of the herniated structure?

- A. Neural tube
- B. Primitive node
- ✓C. Notocord
- D. Neural crest
- E. Primitive streak

Rationale: The intervertebral disc is made up of two parts. The outer annulus fibrosis is derived from mesenchyme, while the inner nucleus pulposus is a remnant of the notocord.

Question #: 19

A 59-year-old man comes to the physician because of painful knees while walking. Physical examination shows limited movement of the knee joint. Which of the following best describes the affected joint?

- A. Considered to be part of the axial skeleton
- B. It is a ball and socket joint
- C. It is an articulation between the head of the radius and the capitulum of the humerus
- ✓D. It is an articulation between the femoral and tibial condyles
- E. It is involved in pronation and supination

Rationale: The Knee joint is a hinge joint that mainly allows for flexion and extension. It is part of the appendicular skeleton. Its articulation is between the femoral condyles and the tibial condyles.

Question #: 20

A 20-year-old man comes to the emergency department because of a 2-hour history of a deep stab wound to his back. Physical examination shows a laceration

that has penetrated the skin, fascia and deep muscles of the back. Which of the following nerve fibers most likely provides direct innervation to the damaged structures?

- A. Ventral rami
- B. Dorsal root
- ✓C. Dorsal rami
- D. Ventral root
- E. Spinal nerve

Rationale: The skin, fascia and deep muscles of the back are supplied by dorsal rami which are branches of the spinal nerve. They are mixed and contains somatic, visceral, sensory and motor fibers.

Question #: 21

A 16-year-old girl is brought to the emergency department by her father because of a 2-hour history of pain in her left elbow after falling from her bike. An x-ray of her elbow shows a fracture of the medial epicondyle. Motor deficits in which of the following movements is most likely to occur if the nerve lying just posterior to the epicondyle is damaged?

- A. Flexion of the index finger
- B. Extension of the wrist
- C. Supination of the forearm
- ✓D. Adduction of the thumb
- E. Abduction of the thumb

Rationale: The nerve that lies posterior to the medial epicondyle is the ulnar nerve. This nerve innervates most of the intrinsic muscles of the hand including the adductor pollicis muscle. This muscle adducts the thumb.

Question #: 22

A 48-year-old woman is admitted to the hospital for surgical repair of a fracture of the spiral groove of the shaft of the humerus. During the surgery, a large hematoma develops from a torn artery. Which of the following will most likely be seen on physical examination because of damage to the nerve that accompanies the torn artery?

- A. Difficulty in flexing the wrist joint
- B. Difficult in flexing the DIP in the little finger
- ✓C. Inability to extend the wrist joint

- D. Inability to extend the elbow joint
- E. Difficulty in laterally rotating the shoulder joint

Rationale: The artery that runs in the spiral groove is the deep brachial artery, and the nerve that accompanies this artery is the radial nerve. If the radial nerve is injured in the spiral groove, all muscles that are innervated distal to this injury are affected, therefore there is no innervation to the muscles of the posterior compartment of the forearm. These muscles are the extensors of the wrist joint. The patient will have wrist drop. The patient will still be able to extend at the elbow joint, because the triceps brachii muscle, which extends the elbow joint receives its innervation, before the radial nerve enters the spiral groove.

Question #: 23

A 48-year-old woman is brought to the emergency department by her husband because she fell on her shoulder after slipping in the shower. Physical examination shows severe swelling and pain at the joint with a visible protrusion of the clavicle. Radiographic studies show shoulder separation. Damage to which of the following structures is most likely responsible for this injury?

- A. Supraspinatus tendon
- B. Coracoacromial ligament
- ✓C. Coracoclavicular ligament
- D. Coracohumeral ligament
- E. Subscapularis tendon

Rationale: Shoulder separation results from damage to the acromioclavicular joint/ligament and the coracoclavicular ligament.

Question #: 24

A 17-year-old boy is brought to the emergency department by his parents because of a 1-hour history of damage to his arm after being thrown from his horse. Radiographic studies show a fracture of the surgical neck of the humerus. Which of the following arteries is most likely damaged due to this injury?

- A. Subscapular
- B. Axillary
- C. Brachial
- D. Deep brachial
- ✓E. Posterior circumflex humeral

Rationale: The axillary nerve and the posterior circumflex humeral artery runs around the surgical neck of the humerus before they pass through the quadrangular space into the posterior shoulder.

Question #: 25

A 70-year-old man with prostate cancer is experiencing sharp shooting pain radiating from his neck into the upper limb. A spinal MRI demonstrates a small metastatic mass in the cervical spine extending into the left intervertebral foramen between C6 and C7. The intervertebral discs appear normal. Which neural structure is most likely being compressed by the metastatic mass to account for the pain?

- A. C8 spinal nerve
- B. Dorsal horn of C6 spinal cord segment
- C. C6 spinal nerve
- D. Dorsal horn of C7 spinal nerve
- ✓E. C7 spinal nerve

Rationale: The correct answer is C7 spinal nerve. The C7 nerve exits above the C7 vertebra.

For the option (Dorsal horn of C7 spinal nerve): dorsal horn is a feature within the spinal cord, not associated with peripheral spinal nerves.

Question #: 26

A 30-year-old man comes to the physician because of a 6-month history of numbness on the anterior chest wall. Physical examination shows decreased touch sensation and sweating over the anterior chest and abdominal wall from the nipple to the umbilicus. Damage to which of the following structures most likely caused the findings in this patient?

- ✓A. Ventral rami
- B. White rami comunicans
- C. Ventral roots
- D. Grey rami communicans
- E. Ventral horn

Rationale: Ventral rami are mixed structures that contain somatic afferents, somatic efferents and postganglionic sympathetics. Ventral rami innervate all parts of the body except the skin over the back and the deep muscles of the back. These are innervated by dorsal rami.

Question #: 27

A 14-year-old boy is brought to the emergency department by his parents because of a 2-hour history of severe pain in his shoulder after falling from a tree. Physical examination shows flattening of his shoulder curvature. Imaging studies show a dislocation of the shoulder joint. Which of the following best describes the affected joint in this patient?

- A. It is an articulation between the capitulum of the humerus with the glenoid cavity
- B. It is a pivot joint
- C. It's movement is restricted to flexion and extension
- D. It is a fibrous joint
- ✓E. It is a ball and socket joint

Rationale: The shoulder joint is a ball and socket joint, with the 'ball' being the head of the humerus and the 'socket' being the glenoid fossa, found on the lateral border of the scapula.

Question #: 28

A 70-year-old woman is brought to the emergency department by the paramedics after she was involved in a motor vehicle collision. Physical examination shows complete paralysis of the upper and lower extremities. An x-ray of the cervical spine shows a posteriorly displaced fracture of the dens which resulting in severe compression of the spinal cord. Which of the following ligaments is most likely torn, resulting in the findings in this patient?

- A. Posterior longitudinal ligaments
- ✓B. Transverse ligament of the atlas
- C. Interspinous ligament
- D. Supraspinous ligament
- E. Nuchal ligament

Rationale: The transverse ligament of the atlas functions to hold the dens against the anterior arch of the axis.

Question #: 29

A 36-year-old man is admitted to the hospital for shoulder surgery to repair a rotator cuff tear. During the procedure, branches of the suprascapular nerve have to be preserved. Which of the following muscles is most likely innervated by this nerve?

- A. Teres major and teres minor muscles
- B. Supraspinatus and subscapularis muscles
- C. Infraspinatus and teres minor muscles
- ✓D. Supraspinatus and infraspinatus muscles
- E. Subscapularis and teres minor muscles

Rationale: The suprascapular nerve innervates the supraspinatus and the infraspinatus muscles.

Question #: 30

A 23-year-old man comes to the emergency department because of a superficial knife wound to the lateral side of his left elbow. Physical examination shows loss of sensation over the lateral dorsum of the hand, but no loss of motor function at the wrist joint. Which of the following nerves is most likely injured in this patient?

- A. Superficial ulnar
- B. Lateral cutaneous of the forearm
- ✓C. Superficial branch of the radial
- D. Radial
- E. Deep branch of the radial

Rationale: Correct: To enter the forearm, the radial nerve travels anterior to the lateral epicondyle of the humerus, through the cubital fossa. The nerve then terminates by dividing into two branches:

Superficial branch (sensory) – contributes to the cutaneous innervation of the dorsal hand and fingers, which is damaged in this patient

Deep branch (motor) – innervates the muscles in the posterior compartment of the forearm, mostly by the posterior interosseous branch. This branch is unaffected because the patient has normal wrist joint movements.

Incorrect: Superficial ulnar nerve will supply cutaneous innervation on the palmar aspect of the medial part of the hand and the 5th and half of the 4th digit. The lateral cutaneous nerve of the forearm (also called lateral antebrachial cutaneous nerve) is a terminal branch of musculocutaneous nerve.